

Mohamed Amine Ezzine

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Research Interests

Gravitational Physics, Theoretical cosmology, Phenomenology of modified gravity, Operator techniques in gravity, Weak field and perturbative approaches.

Education

Master of Science in Physics — Astroparticle Physics Track 2024–2026

Double Degree: KTH Royal Institute of Technology, Stockholm

Relevant coursework: Advanced Quantum Mechanics, General Relativity, Theoretical Particle Physics, Symmetries in Physics.

Laboratory work: Analysis of real Higgs boson data using the ROOT framework, with emphasis on visualization and statistical inference.

Master's thesis: *Time Delay in Bimetric Gravity: From the Eikonal Approximation to the Wigner–Smith Operator.*

Bachelor of Science in Mathematics, Physics, and Engineering Sciences 2020–2023

IPEST, Tunis, Tunisia & ISAE-SUPAERO, Toulouse, France

Relevant coursework: Linear Algebra, Vector Calculus, Partial Differential Equations, Classical Electromagnetism, Fluid Dynamics.

Engineering Degree in Aerospace Engineering 2023–2025

Double Degree: ISAE-SUPAERO, Toulouse, France

Relevant coursework: Statistical Physics, Real Analysis, Partial Differential Equations, Numerical Methods, Non-Equilibrium Dynamics.

Research Experience

Master's Thesis: Time Delay Calculations in Bimetric Gravity 2025–2026

Supervisor: Dr. Fawad Hassan

- **Research question:** How can scattering-amplitude techniques be applied to gravitational interactions in bimetric gravity, and which classical observables can be extracted from this quantum description?
- **Investigation:** Diagonalized the bimetric gravity Lagrangian in the weak-field and proportional background limits into massive and massless graviton modes and Computed gravitational scattering amplitudes in the high-energy regime using the eikonal approximation. Extracted time-delay observables from the scattering matrix using stationary-phase methods and explored operator-based approaches using the Wigner–Smith operator.
- **Results:** Derived corrections to General Relativity predictions in the weak-field limit and reproduced literature constraints on bimetric gravity parameters using Cassini mission time-delay observations. This work is currently being developed into a short research manuscript; draft available upon request.
- **Skills acquired:** This work developed tools in scattering theory, Feynman diagrams resummation techniques, and translating scattering amplitudes into observational predictions.

Research Project: Operator Techniques in Electromagnetic Scattering 2023–2024

Supervisor: Dr. Valentin Maziers

- **Research question:** Can operator-based methods, particularly the Wigner–Smith time-delay operator, be used to engineer electromagnetic wave states with minimal propagation delay and energy loss in complex media?
- **Investigation:** Developed a simplified scattering model with multiple input and output channels and implemented an FDTD simulation to extract scattering matrix of an electromagnetic wave propagating in the scattering system. Computed eigenstates of the Wigner–Smith operator under perturbations of the medium properties and studied both single-parameter and multi-parameter variations.
- **Results:** Showed that the eigenstates of the Wigner–Smith operator correspond to wave states with minimal propagation delay and reduced energy loss. Demonstrated that generalized operator perturbations localize energy in regions where medium parameters are modified, and identified interference effects arising from simultaneous variations of multiple parameters.
- **Skills acquired:** This work developed tools in computational physics both through algorithm development and use of software.

Research Project: Non-relativistic Models of Nuclear Interactions

2024–2025

Independent work

- **Research question:** How do renormalization and analytic properties emerge in the Chew–Low model as a simplified framework for interacting quantum fields?
- **Investigation:** Studied the renormalization structure of the Chew–Low model and computed scattering amplitudes within the one-meson one-nucleon sector. Analyzed the analytic structure of the amplitudes and explored the role of dispersion relations in connecting physical observables to amplitude singularities.
- **Results:** Developed familiarity with perturbative renormalization techniques, scattering theory, and dispersive methods in quantum field theory.
- **Skills acquired:** This work developed tools in quantum field theory scattering theory, including S-matrix methods and dispersion relations.

Advanced Independent Study

- **Quantum Field Theory:** Independent study of canonical quantization, scattering theory, and introductory axiomatic QFT.
- **General Relativity:** Complementing university lectures on general relativity by a self-study of space-time and geometry to gain a better understanding of coordinate independent methods such as differential forms, and exterior derivatives.
- **Differential Geometry (ongoing):** Independent study of smooth manifolds and geometric methods relevant to modern theoretical physics.

Technical Skills

- **Programming:** Python (NumPy, SciPy, Matplotlib), MATLAB (numerical ODEs/PDEs), Mathematica, C, ROOT.
- **Typesetting:** L^AT_EX.

Languages

English (fluent) French (bilingual) Arabic (native) Swedish (basic)

Scientific Outreach and Activities

- Member, ISAE-SUPAERO Debate Association.

- Contributor, *Our Giant Leap* podcast — Space Generation Advisory Council (Diversity and Gender Equality Project Group).